

Develop a program that redirects standard input and Output using file descriptors and dup2(). Demonstrate how shells implement 1/0 redirection.

Code:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int file;
    int original_stdout;

    /* -----
       OUTPUT REDIRECTION
       ----- */

    printf("Starting output redirection...\n");

    /* Save original stdout */
    original_stdout = dup(STDOUT_FILENO);

    /* Open output file */
    file = open("output.txt",
                O_WRONLY | O_CREAT | O_TRUNC,
                0644);

    if (file == -1)
    {
```

```
        perror("open");
        return 1;
    }

    /* Redirect stdout to file */
    dup2(file, STDOUT_FILENO);

    close(file);

    /* This goes into output.txt */
    printf("This line is redirected to the file.\n");
    printf("Hello from dup2()!\n");

    /* Restore stdout */
    dup2(original_stdout, STDOUT_FILENO);

    close(original_stdout);

    printf("Output redirection finished.\n");

    /* -----
       INPUT REDIRECTION
       ----- */
```

```

    printf("\nStarting input redirection...\n");

    file = open("input.txt", O_RDONLY);

    if (file == -1)
    {
        perror("input.txt");
        return 1;
    }

    /* Redirect stdin to input.txt */
    dup2(file, STDIN_FILENO);

    close(file);

    char text[100];

    /* fgets now reads from input.txt */
    if (fgets(text, sizeof(text), stdin) != NULL)
    {
        printf("Data read from redirected input:\n");
        printf("%s", text);
    }

    printf("\nInput redirection finished.\n");

```

```

    return 0;
}

```

Output:

```

geya@LenovoL480:~/OS/Practical/Week 9$ ./task2
Starting output redirection...
Output redirection finished.

Starting input redirection...
Data read from redirected input:
This is a test file for the file copy program.

Input redirection finished.
geya@LenovoL480:~/OS/Practical/Week 9$ ls
copy_low.txt  copy_standard.txt  input.txt  output.txt  task1  task1.c  task2  task2.c
geya@LenovoL480:~/OS/Practical/Week 9$ |

```